

2 SP-201 Red Multi-Agent Research & Fact-Checking Pipeline

CS 4850 - Section 02 – Spring 2026

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Team Members:

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Collaboration Tools:

Communication	—	Teams meetings, Cellphones (Call/Text)
Collaboration	—	Teams (in between weekly status meetings on Teams !)

Abstract:

The design and implementation of a multi-agent autonomous research and fact-checking system using existing big language models and agent orchestration frameworks is the main goal of this project. In order to generate a verified, credited research report, the system is built to process complicated, open-ended research prompts by breaking them down into structured subtasks and coordinating several specialized agents.

A central Manager Agent evaluates the user prompt and assigns tasks to several worker agents in the architecture's Manager–Worker pattern. A search agent uses free, publicly available APIs to retrieve pertinent data from other data sources. A Fact-Checker Agent cross-references assertions with the original sources to identify discrepancies and lessen hallucinations, while a Synthesizer Agent incorporates the collected data into a logical draft. The project's scope encompasses multi-agent coordination, session state management, tool-based external data retrieval, and hierarchical agent design. The finished system seeks to show how autonomous agents can work together to enhance research reproducibility, transparency, and reliability in AI-assisted knowledge creation.

Platform:

Frontend: React and ASP.NET

Backend: Nodejs + Google ADK

STATEMENT OF PARTICIPATION

By submitting this assignment, I acknowledge that I will participate in all meetings, communications, deliverables, and other tasks necessary to complete the project. If I do not participate, I understand that Professor Perry will meet with me to remedy the situation.